

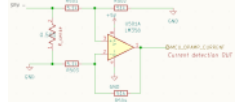
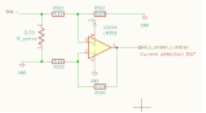
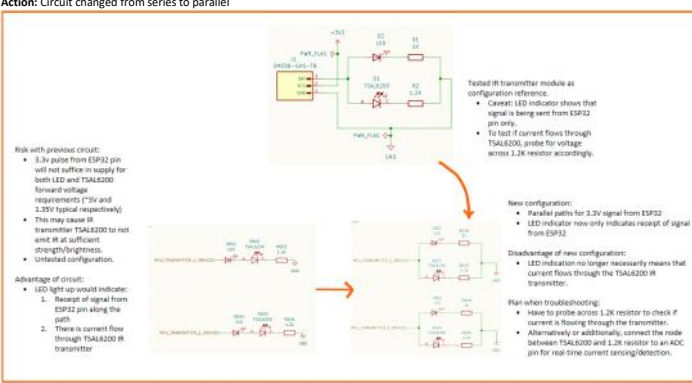
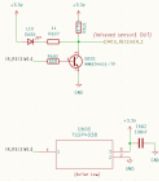
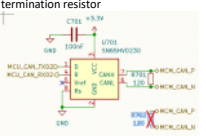
Method and objective:

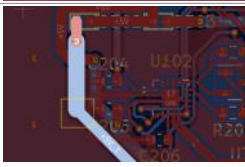
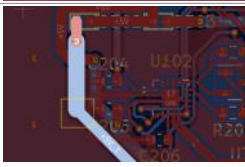
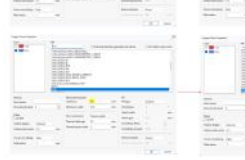
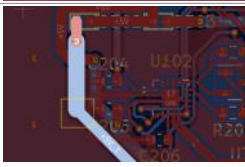
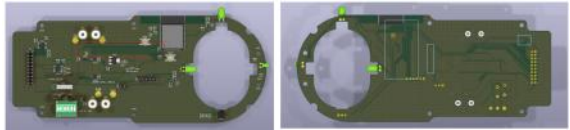
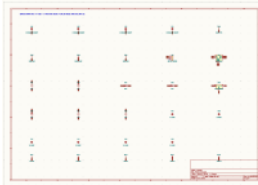
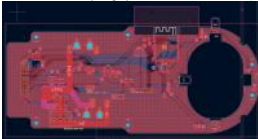
- 1. Have GPT6 review existing design and output list of recommended changes. We verify and implement changes accordingly. To see what errors and potential functionality risks might have been overlooked.
- 2. Have GPT6 edit latest updated design (after step 1) on KICAD, to see what sort of improvements it would implement, and the quality of its execution
- 3. Have GPT6 completely redesign the Rdrop board based on provided description of project requirements and limitations, and generate fabrication files. To see how effective it is in generating a project from scratch.

Prompts:

Step 1	The Rdrop project is a rat bait dropping device that can be attached to the bottom of the drone. 2 pairs of IR transmitter and receiver are used in my current set up, one digital and one analog-based. My constraints are as follows: - As in the Butiks (1).DXF file, that is the bottom mechanism (where the turning gears are placed etc), above which will sit the bottom of the PCB. - As you can access in the current rdrop PCB file, the edge cuts are according to the size constraints of the top casing/cover of the device that it would be mounted on. There is a hole that have followed the sizing of the larger of 2 rat bait types that the project would use and the hole would allow the bait to fall through and the IR sensors would work accordingly. - The finalised connector that would connect to the drone is the JAE MX23A type (18pin standard) which would bring across the drone battery voltage that would be stepped down through either an APM (Aonic Power Module) or an external, solderable UBEC regulator. It serves the feature of Selectable servo input voltage (+7.4V / +5V) - Servo input voltage is now configurable from the drone battery input via one of two regulator/fuse paths: - For the APM: 18S / 78V would be stepped down to 5V and then go to the respective paths and the 5V would also feed +5V directly to the servo feed (SRV+) - For the UBEC: drone battery voltage would be stepped down to 7.4V and then into LM1084-5.0 Regulator (APM power path would bypass this). - The rdrop board is targeted to carry at least 15A of current, taking into account current spikes when using the servo that could reach up to 7A when 7.4V is used to feed it. Taking into account miscellaneous board component usage that had been surveyed to be up to 2-3A max at normal operation usage. Going through the files I've given you access to, and permission to scour the web for the relevant datasheets of the component, understand my design, and verify whether it would serve my intended design and functionality accordingly. Otherwise, list all the corrections I would need to do, direct reason as to why it is wrong and why your correction is right, and also listing any parts that might need flagging, if any. My aim is to have a functional board first, any refinement we discuss later. Estimate as well with my set up and in your recommended corrections, the current consumption of the board overall, and break down for each component based on their data sheet. Do not edit the original files but instead create a duplicate adjacent file and edit there
Step 2	Now, I want to evaluate performance of gpt6 in editing and refining pcb on its own, and compare with the design that i have. In the given Rdrop Astra folder, edit my pcb on how you would refine its layout, and traces and planes, adhering to schematic and functionality by making a new folder in the Rdrop Astra folder called "GPT6 Refined", copy the relevant kicad files and make edits via kicad 9 on the computer. and generate the references images with pcb layout and 3 layout compared to the original one. list the changes made.. Ask me any information you might need to know on the target specs etc for this 2 layer 1oz board. board sizing 15A support. Maintain the shape and cutout and the components. Positions of magnetic sensor, IR transmitter and receivers must also remain. The mounted connectors must also remain in their positions.
Step 3	Act as an expert hardware engineer. Redesign a complete 2-layer PCB project for this Rdrop project, from schematic to pcb to 3d model and fabrication files, for an ESP32 wroom 32e-based rat bait drop attachment to a drone. Include the functional blocks such as in the original Rdrop design: (1) 7.4V to 5V LDO, 5V to 3.3V step-down regulator using an AMS1117 with input/output decoupling caps, (2) ESP32-WROOM-32 module, (3) ASS5600 magnetic sensor (4) 1 pair of TSAL6200 and TEFT4300 and 1 more pair of TSAL6200 and TSSP4038, a mini mum 16 pin connector, board must support up to 15A, a 3pin servo connector with 2 fuses (1 soldered at a time) that trips at 15A, to deliver either 7.4 V or 5V to the servo, standoffs and UBEC input pad (regulator wires would be soldered onto them) must remain for the external regulator choices (must also have a 3pin header for UBEC output). You may experiment with any better alternatives of the components apart from the ones that had been selected in the previous design. Use exact component part numbers available in standard supply chains (e.g., DigKey/LCSC). Use 0603 size for all standard resistors and capacitors. Ensure all components have valid, standard KiCad library footprints assigned to prevent schematic-to-layout parity errors The finished board must be within the outline size and cutout of the original, with mounting holes at the corner. Open KiCad, generate the multi-sheet schematic, perform component placement, and route all copper traces. Run both Electrical Rule Check (ERC) and Design Rule Check (DRC) iteratively until zero errors or unresolved net parity conflicts remain. Export the final Gerber files, Excellon drill files, BOM, and CPL (pick-and-place) data. Save and organise files into a new folder in the Rdrop Astra folder called "GPT6 New" Generate also a report of your new design (website form)

Findings:

Step 1	GPT6 Output Review file that contains list of recommended changes  READ_ME_REVIEW	Recommendation/Risk highlighted 1) UBEC Max 14S 2) Make the 5 V / 7.4 V selection mutually exclusive 3) Replace the servo shunt and redesign the current measurement  4) Qualify the entire 15 A path, including fuses and connectors 5) Connect ASS5600 correctly and add iPC pull-ups 6) Drive the IR emitters independently of the visible LEDs	Action taken after verifying? No action. Acknowledged, as the UBEC would only be used for 14S drone batteries No action. This has been considered in design and usage, where only one of the fuses would be soldered on at a time, depending on which power regulator is used. Additionally there are Schottky diodes in place to protect from backflow of current. Action: Shunt and resistor values have been updated.  Action: Quick run through and double check. Fuse can handle expected current draw, J1 and J2 plane size supports 15A, standoffs are 3 0A rated, JAE double checked its suitability. Action: 5V shorted accordingly and added pull-ups to iPC SDA SCL Action: Circuit changed from series to parallel  Action: Moved LED to near Q601  No action. TEFT4300 circuit has been revisited extensively and the current setup has been tested and can function as intended. Action: R702 removed and marked as DNP (Do Not Populate) but remain in cases where the other device does not have its own built in CAN termination resistor  No action. This is a refined verification and time constraints to not permit it, but regulator choice had been reviewed and approved by Senior Embedded Engineer previously. No action. Passed previous review. No action. Passed previous review and such concern was not raised then. This is an added layer of security, where the planes are designed to support current expected, and fuse is present at critical points of power draw elsewhere.				
Step 2	Website generated containing PCB layouts and 3D comparison Detailed change report (.md and pdf)  Rdrop Astra - GPT6 Re... Verification Records  Rdrop Astra	Review, analyze and comment on changes made Comments from Change Report, Detailed Changes and Verification Records pdf: <table><tr><th>Change made</th><th>Comment</th></tr><tr><td>GND clearance expansion</td><td>Good that it reduces conservativity of previous clearance, to expand return path and remove isolated thermal-spokes</td></tr></table>	Change made	Comment	GND clearance expansion	Good that it reduces conservativity of previous clearance, to expand return path and remove isolated thermal-spokes	Screen capture of change 
Change made	Comment						
GND clearance expansion	Good that it reduces conservativity of previous clearance, to expand return path and remove isolated thermal-spokes						

 - Rdrop Astra - Verificat... - Detailed Changes  - Rdrop Astra - Detailed... "GPT6 Refined" folder of copied and edited KICAD project files	<table border="1"> <tr> <td data-bbox="319 280 456 418">GND stitching vias</td><td data-bbox="456 280 722 418">Quite random placement even over serial barcode region, but good initiative to stitch wide via-less areas of top and bottom copper pours</td><td data-bbox="729 280 989 418">  </td></tr> <tr> <td data-bbox="319 418 456 591">+5V_1, +5V_2 and +UBEC_OUT trace expansion</td><td data-bbox="456 418 722 591">Good initiative but expansions were mid-trace, abrupt (no taper although low risk of acid traps), and not throughout the entire trace. Initially 1mm considered to suffice for 5V delivery with relatively low power draw</td><td data-bbox="729 418 989 591">  </td></tr> </table> Overall comment: Changes are minimal, conservative and only in copper pour/trace sizing and relief. Good initiatives especially on taking liberty with GND pour clearances, but execution can be more complete (inconsistent track size) and thoughtful (via placement). Careful in implementing changes of which verification method and records are provided. No changes or optimizations in component and track placements.	GND stitching vias	Quite random placement even over serial barcode region, but good initiative to stitch wide via-less areas of top and bottom copper pours		+5V_1, +5V_2 and +UBEC_OUT trace expansion	Good initiative but expansions were mid-trace, abrupt (no taper although low risk of acid traps), and not throughout the entire trace. Initially 1mm considered to suffice for 5V delivery with relatively low power draw		
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Step 3 Website detailing new design	Comments at a glance: - Needs extensive validation of circuit and inter-circuit design and functionality, and component placement. - Good tracing at some parts (straight one-way horizontal traces on top copper), but some parts are clearly wonky most likely due to the workaround feature used  - Good added layers of gates to improve logic efficiency e.g. individual drivers for IR transmitters - Placement of connectors requires more user-provided constraint and guidance so they are more sensible in terms of product integration, outcome and use - Efficient use of space could be in place if it were free to design the shape (not constrained by existing board outline) - Smaller BOM: The new Rdrop has 69 PCB footprints versus 78 in the original—9 fewer (about 12%), counting mounting holes, test points, and other board features. - Schematic generated is not human-readable without extensive zooming  - PCB looks passable and decent and can be good with refinements - Maximised via use which can be good - No use of power planes - not so efficient but perhaps can be encouraged in prompt - Instead, extremely large power traces were used (4.5mm) - unnecessarily overpowered, use planes instead to reduce loop inductance and noise 	- Good that it can generate fabrication files too - can automate job - It also generates harness/connection list which is an efficient to keep track of traces and troubleshoot in the future - Could also look into it: integration in workflow in future						